Self-consistent optimal-transport formulations of two squared GW problems

High-level account

Let

$$\mathcal{J}(\pi) = \iint \left(s_X(x,x') - s_Y(y,y')\right)^2 d\pi(x,y)\, d\pi(x',y')$$

be the squared Gromov–Wasserstein objective over $\pi \in \Pi(\mu,\nu)$. At a minimizer $\pi_\star$, the first variation of this quadratic functional must be nonnegative in every feasible direction. Consequently, $\pi_\star$ solves an ordinary Kantorovich problem whose cost is the linearization of $\mathcal{J}$ at $\pi_\star$. Thus GW can be written as the fixed-point condition

$$\boxed{\pi_\star \in \mathrm{OT}_{c_{\Theta(\pi_\star)}}(\mu,\nu),}$$

where the cost parameter $\Theta(\pi_\star)$ is itself a moment matrix of $\pi_\star$.

For the inner-product structures

$$s_X(x,x') = x^\top x', \qquad s_Y(y,y') = y^\top y',$$

the relevant matrix is

$$M_\pi = \mathbb{E}_\pi[XY^\top],$$

and an optimal coupling satisfies

$$\pi_\star \in \underset{\gamma\in\Pi(\mu,\nu)}{\arg\max}\, \mathbb{E}_\gamma[X^\top M_{\pi_\star} Y].$$

The full GW objective depends on π only through M_π. Hence, when both marginals are Gaussian, any optimal cross-moment can be realized by a jointly Gaussian coupling. In equal dimension, with positive-definite centered Gaussian marginals, every optimizer is in fact a deterministic linear Gaussian coupling.

For squared Euclidean structures

$$s_X(x,x') = \left\|x - x'\right\|^2, \qquad s_Y(y,y') = \left\|y - y'\right\|^2,$$

one first lifts x to $(\left\|x\right\|^2, x, 1)$. The same matrix fixed-point statement then holds in the lifted space. For centered marginals it reduces to

$$\pi_\star \in \underset{\gamma\in\Pi(\mu,\nu)}{\arg\max}\, \mathbb{E}_\gamma\left[\left\|X\right\|^2 \left\|Y\right\|^2 + 4X^\top K_{\pi_\star} Y\right], \qquad K_\pi = \mathbb{E}_\pi[XY^\top].$$

Unlike the inner-product case, the radial fourth-order term $\mathbb{E}[\left\|X\right\|^2 \left\|Y\right\|^2]$ is not determined by the cross-covariance. This permits non-Gaussian dependence to outperform every Gaussian coupling. An explicit example is given below for $\mathcal{N}(0,1)$ and $\mathcal{N}(0,I_{10})$.

1 The general linearization principle

Let $\mu \in \mathcal{P}(\mathbb{R}^d)$ and $\nu \in \mathcal{P}(\mathbb{R}^e)$ have enough moments for all expressions below to be finite. Let s_X and s_Y be symmetric structure functions, and define

$$L\big((x,y),(x',y')\big) = \big(s_X(x,x') - s_Y(y,y')\big)^2.$$

For $\pi \in \Pi(\mu,\nu)$, set

$$\mathcal{J}(\pi) = \iint L(z,z')\,d\pi(z)\,d\pi(z'), \qquad \ell_\pi(z) = \int L(z,z')\,d\pi(z').$$

Proposition 1 (Self-consistent OT condition)**.** *If $\pi_\star$ minimizes $\mathcal{J}$ over $\Pi(\mu,\nu)$, then*

$$\pi_\star \in \operatorname*{arg\,min}_{\gamma\in\Pi(\mu,\nu)} \int \ell_{\pi_\star}(x,y)\,d\gamma(x,y).$$

Thus every GW minimizer is an OT minimizer for the cost $c_{\pi_\star} = \ell_{\pi_\star}$ generated by that same coupling.

Proof. For any $\gamma \in \Pi(\mu,\nu)$, put

$$\pi_t = (1-t)\pi_\star + t\gamma.$$

Since $\pi_\star$ is a minimizer,

$$0 \le \frac{d}{dt}\mathcal{J}(\pi_t)\bigg|_{t=0^+} = 2\int \ell_{\pi_\star}(z)\,d(\gamma-\pi_\star)(z).$$

Rearranging gives the claimed Kantorovich optimality condition. □

The condition is necessary; by itself it is not generally sufficient for global GW optimality, because it is a nonlinear fixed-point condition.

2 Bilinear feature kernels and the matrix parameter

Suppose that the two structures admit finite-dimensional bilinear representations

$$s_X(x,x') = \phi(x)^\top A\phi(x'), \qquad s_Y(y,y') = \psi(y)^\top B\psi(y'),$$

where A and B are symmetric matrices. Define the cross-feature moment

$$M_\pi = \int \phi(x)\psi(y)^\top\,d\pi(x,y).$$

Independence of the two copies drawn from π gives

$$\iint s_X(x,x')s_Y(y,y')\,d\pi\,d\pi = \operatorname{tr}\Big(AM_\pi BM_\pi^\top\Big).$$

Therefore

$$\mathcal{J}(\pi) = C_{\mu,\nu} - 2\operatorname{tr}\Big(AM_\pi BM_\pi^\top\Big),$$

where $C_{\mu,\nu}$ depends only on the marginals.

Moreover, after deleting a function of x and a function of y, which do not affect a Kantorovich minimizer,

$$\ell_\pi(x,y) \equiv_{\mu,\nu} -2\phi(x)^\top AM_\pi B\psi(y).$$

Here $c \equiv_{\mu,\nu} \widetilde{c}$ means

$$c(x,y) - \widetilde{c}(x,y) = f(x) + g(y)$$

for some f and g. Such terms have constant integral over $\Pi(\mu,\nu)$.

Consequently, with

$$\Theta_\pi = AM_\pi B,$$

every GW optimizer obeys

$$\boxed{\pi_\star \in \operatorname*{arg\,min}_{\gamma\in\Pi(\mu,\nu)} \int -\phi(x)^\top \Theta_{\pi_\star} \psi(y)\, d\gamma(x,y).}$$

This is the desired ordinary OT problem with a matrix-valued parameter that is determined self-consistently by the optimal coupling.

3 Inner-product GW and Gaussian couplings

Take

$$s_X(x,x') = x^\top x', \qquad s_Y(y,y') = y^\top y'.$$

Then $\phi(x) = x$, $\psi(y) = y$, $A = I_d$, $B = I_e$, and

$$M_\pi = \mathbb{E}_\pi[XY^\top].$$

The objective and its self-consistent OT formulation are

$$\mathcal{J}_{\mathrm{ip}}(\pi) = C_{\mu,\nu} - 2\,\|M_\pi\|_{\mathrm{F}}^2\,,$$
$$\pi_\star \in \operatorname*{arg\,max}_{\gamma\in\Pi(\mu,\nu)} \int x^\top M_{\pi_\star} y\, d\gamma(x,y).$$

Thus inner-product GW maximizes the squared Frobenius norm of a second cross-moment.

3.1 Why a Gaussian optimizer exists

Let

$$\mu = \mathcal{N}(m_0,\Sigma_0), \qquad \nu = \mathcal{N}(m_1,\Sigma_1).$$

For any coupling π, let

$$K_\pi = \mathrm{Cov}_\pi(X,Y), \qquad M_\pi = m_0 m_1^\top + K_\pi.$$

Because K_π comes from an actual coupling, the block covariance matrix

$$\Sigma_\pi = \begin{pmatrix} \Sigma_0 & K_\pi \\ K_\pi^\top & \Sigma_1 \end{pmatrix}$$

is positive semidefinite. Hence the jointly Gaussian law

$$\widehat{\pi} = \mathcal{N}\left(\begin{pmatrix} m_0 \\ m_1 \end{pmatrix}, \Sigma_\pi\right)$$

is a coupling of μ and ν with exactly the same M_π.

Since $\mathcal{J}_{\mathrm{ip}}$ depends on π only through M_π,

$$\mathcal{J}_{\mathrm{ip}}(\widehat{\pi}) = \mathcal{J}_{\mathrm{ip}}(\pi).$$

Applying this replacement to any optimizer proves that a jointly Gaussian inner-product GW optimizer always exists.

The fixed-point OT formula gives the same conclusion more geometrically. If $M_{\pi_\star}$ is invertible, set

$$z = M_{\pi_\star}^\top x.$$

Then maximizing

$$x^\top M_{\pi_\star} y = z^\top y$$

is equivalent, up to marginal terms, to minimizing $\|z-y\|^2$, since

$$-z^\top y = \frac{1}{2}\|z-y\|^2 - \frac{1}{2}\|z\|^2 - \frac{1}{2}\|y\|^2.$$

The quadratic OT map between two nondegenerate Gaussian laws is affine. Composing it with $x \mapsto M_{\pi_\star}^\top x$ makes Y an affine function of X, so the optimal joint law is Gaussian.

3.2 Equal-dimensional centered case: every optimizer is Gaussian

Assume now

$$\mu = \mathcal{N}(0, \Sigma_0), \qquad \nu = \mathcal{N}(0, \Sigma_1), \qquad \Sigma_0, \Sigma_1 \succ 0,$$

with both measures on $\mathbb{R}^d$. A matrix $K = \mathbb{E}[XY^\top]$ is feasible exactly when it can be written as

$$K = \Sigma_0^{1/2} C \Sigma_1^{1/2}, \qquad \|C\|_{\mathrm{op}} \le 1.$$

Hence the moment problem is

$$\max_{\|C\|_{\mathrm{op}} \le 1} \operatorname{tr}\Big(\Sigma_0 C \Sigma_1 C^\top\Big).$$

Let the eigenvalues of Σ_0 and Σ_1 be

$$\alpha_1 \ge \cdots \ge \alpha_d > 0, \qquad \beta_1 \ge \cdots \ge \beta_d > 0.$$

The trace rearrangement inequality gives

$$\max_{\|C\|_{\mathrm{op}} \le 1} \operatorname{tr}\Big(\Sigma_0 C \Sigma_1 C^\top\Big) = \sum_{i=1}^d \alpha_i \beta_i.$$

The maximum is attained by an orthogonal $C_\star$ aligning the ordered eigenspaces, with arbitrary signs inside one-dimensional eigenspaces and arbitrary orthogonal choices inside repeated eigenspaces. Since both covariances are positive definite, every maximizing contraction saturates all directions and is orthogonal on $\mathbb{R}^d$.

For such a maximizer,

$$K_\star = \Sigma_0^{1/2} C_\star \Sigma_1^{1/2}$$

and

$$\Sigma_1 - K_\star^\top \Sigma_0^{-1} K_\star = \Sigma_1^{1/2}(I - C_\star^\top C_\star)\Sigma_1^{1/2} = 0.$$

Therefore every coupling having this optimal cross-covariance satisfies

$$Y = K_\star^\top \Sigma_0^{-1} X \qquad \text{almost surely.}$$

It is a deterministic linear image of a Gaussian variable and is therefore a jointly Gaussian coupling. Thus, in this nondegenerate equal-dimensional centered setting, not only does a Gaussian optimizer exist: every optimizer is Gaussian.

4 Squared Euclidean structures via a quadratic lift

Now take

$$s_X(x, x') = \left\|x - x'\right\|^2, \qquad s_Y(y, y') = \left\|y - y'\right\|^2.$$

Introduce

$$\Phi_d(x) = \begin{pmatrix} \|x\|^2 \\ x \\ 1 \end{pmatrix}, \qquad H_d = \begin{pmatrix} 0 & 0 & 1 \\ 0 & -2I_d & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

Then

$$\left\|x - x'\right\|^2 = \Phi_d(x)^\top H_d \Phi_d(x').$$

Likewise define Φ_e and H_e for y. With the lifted cross-moment

$$\widetilde{M}_\pi = \mathbb{E}_\pi\left[\Phi_d(X)\Phi_e(Y)^\top\right],$$

the general bilinear-feature calculation yields

$$\Theta_\pi^{\text{dist}} = H_d \widetilde{M}_\pi H_e$$

and

$$\boxed{\pi_\star \in \underset{\gamma \in \Pi(\mu,\nu)}{\arg\min} \int -\Phi_d(x)^\top \Theta_{\pi_\star}^{\text{dist}} \Phi_e(y)\, d\gamma(x, y).}$$

Thus squared-distance GW also has a matrix-parameterized OT fixed point, but the matrix acts on quadratic lifts rather than on x and y alone.

4.1 Centered simplification

Assume $\mathbb{E}[X] = \mathbb{E}[Y] = 0$ and write

$$K_\pi = \mathbb{E}_\pi[XY^\top].$$

For fixed (x, y), define

$$h_\pi(x, y) = \mathbb{E}_\pi\left[\left\|x - X'\right\|^2 \left\|y - Y'\right\|^2\right].$$

Expanding the product and deleting marginal-only terms gives

$$h_\pi(x, y) \equiv_{\mu,\nu} \|x\|^2 \|y\|^2 + 4x^\top K_\pi y.$$

Since the linearized squared-discrepancy cost is

$$f(x) + g(y) - 2h_\pi(x, y),$$

every optimizer satisfies

$$\boxed{\pi_\star \in \arg\max_{\gamma\in\Pi(\mu,\nu)} \int \left(\|x\|^2 \|y\|^2 + 4x^\top K_{\pi_\star} y\right) \, d\gamma(x,y).}$$

This is an ordinary OT problem with polynomial cost

$$c_{K_\star}(x,y) = -\|x\|^2 \|y\|^2 - 4x^\top K_\star y, \qquad K_\star = \mathbb{E}_{\pi_\star}[XY^\top].$$

There is also a useful global form. Let (X',Y') be an independent copy of (X,Y) and set

$$a = \mathbb{E}\|X\|^2, \qquad b = \mathbb{E}\|Y\|^2, \qquad q_\pi = \mathbb{E}_\pi[\|X\|^2 \|Y\|^2].$$

A direct expansion gives

$$\mathbb{E}\Big[\|X-X'\|^2 \|Y-Y'\|^2\Big] = 2q_\pi + 2ab + 4\|K_\pi\|_{\mathrm{F}}^2 .$$

The other terms in the squared GW discrepancy depend only on the marginals. Therefore minimizing squared-distance GW is equivalent to maximizing

$$\boxed{\mathcal{F}(\pi) = q_\pi + 2\|K_\pi\|_{\mathrm{F}}^2 .}$$

The new quantity q_π is a mixed fourth-order radial moment. This is the term that breaks Gaussian closure.

5 Counterexample: Gaussian marginals, but no Gaussian optimizer

Let

$$\mu = \mathcal{N}(0,1) \quad \text{on } \mathbb{R}, \qquad \nu = \mathcal{N}(0, I_{10}) \quad \text{on } \mathbb{R}^{10}.$$

5.1 Best value attainable by a jointly Gaussian coupling

For a jointly Gaussian coupling, write

$$k = \mathbb{E}[XY^\top] \in \mathbb{R}^{1\times 10}.$$

Positive semidefiniteness of the block covariance gives

$$\|k\|^2 \le 1.$$

By Isserlis' formula,

$$q_\pi = \mathbb{E}[X^2 \|Y\|^2] = \sum_{j=1}^{10} \mathbb{E}[X^2 Y_j^2] = 10 + 2\|k\|^2 .$$

Consequently every jointly Gaussian coupling satisfies

$$\mathcal{F}(\pi) = q_\pi + 2\|k\|^2 = 10 + 4\|k\|^2 \le 14.$$

The value 14 is attained, for example, by taking $Y_1 = X$ and making $Y_2, \ldots, Y_{10}$ independent standard Gaussians.

5.2 A better non-Gaussian coupling

Draw

$$Y \sim \mathcal{N}(0, I_{10})$$

and put

$$S = \|Y\|^2 \sim \chi^2_{10}.$$

Let F_{10} and F_1 denote the distribution functions of χ^2_{10} and χ^2_1. Define

$$U = F_{10}(S), \qquad A = F_1^{-1}(U).$$

Then U is uniform on $(0,1)$ and $A \sim \chi^2_1$. Let ε be an independent Rademacher variable and set

$$X = \varepsilon\sqrt{A}.$$

It follows that $X \sim \mathcal{N}(0,1)$, so this is a valid coupling of the two Gaussian marginals. Moreover,

$$\mathbb{E}[X \mid Y] = 0, \qquad k = \mathbb{E}[XY^\top] = 0.$$

On the other hand, the squared radii are comonotonically coupled, and hence

$$q_\pi = \mathbb{E}[AS] = \int_0^1 F_1^{-1}(u) F_{10}^{-1}(u)\, du = 15.85744810\ldots.$$

Thus

$$\mathcal{F}(\pi) = 15.85744810\ldots > 14.$$

Because larger $\mathcal{F}$ means smaller squared-distance GW objective, this non-Gaussian coupling is strictly better than every jointly Gaussian coupling. Therefore no Gaussian coupling can be optimal for this pair of Gaussian marginals.

The joint law is non-Gaussian: X^2 is a deterministic increasing function of $\|Y\|^2$, while

$$\mathrm{Cov}(X, Y) = 0.$$

Under joint Gaussianity, zero cross-covariance would imply independence, which is impossible here.

5.3 Why the phenomenon is not a numerical accident

For general n, define

$$I_{1,n} = \int_0^1 F_1^{-1}(u) F_n^{-1}(u)\, du.$$

The standardized chi-square quantiles satisfy, in $L^2(0,1)$,

$$\frac{F_n^{-1}(u) - n}{\sqrt{2n}} \longrightarrow \Phi^{-1}(u),$$

where Φ is the standard normal distribution function. Hence

$$I_{1,n} = n + \sqrt{2n}\, c + o(\sqrt{n}), \qquad c = \int_0^1 F_1^{-1}(u)\Phi^{-1}(u)\, du > 0.$$

The positivity follows because both quantile functions are strictly increasing and nonconstant. Therefore

$$I_{1,n} > n + 4$$

for all sufficiently large n, while the best jointly Gaussian value is exactly $n+4$. The counterexample is therefore structural: the radial mixed fourth moment can gain an order-$\sqrt{n}$ amount that is invisible to Gaussian cross-covariance optimization.

Remark 1 (Same ambient dimension and nondegenerate covariances)**.** *The preceding example uses different intrinsic dimensions, which is natural for GW. It can also be converted into a same-dimensional full-rank example. On* $\mathbb{R}^{10}$*, take*

$$\mu = \mathcal{N}(0, I_{10}), \qquad \nu_\delta = \mathcal{N}(0, \operatorname{diag}(1, \delta, \ldots, \delta)) .$$

For every jointly Gaussian coupling,

$$\mathcal{F}(\pi) \leq 14(1 + 9\delta).$$

Indeed,

$$\operatorname{tr}(\operatorname{Cov}(Y)) = 1 + 9\delta$$

and the maximal squared Frobenius norm of a feasible cross-covariance is also $1 + 9\delta$*.*

For a non-Gaussian competitor, couple $\|X\|^2 \sim \chi^2_{10}$ *comonotonically with* $\|Y\|^2$*, and choose the direction of* X *independently and uniformly on the sphere. This gives* $K_\pi = 0$*. As* $\delta \downarrow 0$*, its value of* $\mathcal{F}$ *converges to*

$$I_{1,10} = 15.85744810\ldots,$$

whereas the Gaussian upper bound converges to 14*. Hence the strict inequality persists for all sufficiently small* $\delta > 0$*. Both Gaussian marginals can therefore be chosen nondegenerate and supported on the same Euclidean space.*

Conclusion

Both squared GW problems admit the same formal description:

$$\pi_\star \in \mathrm{OT}_{c_{\Theta(\pi_\star)}}(\mu, \nu).$$

For inner products,

$$\Theta(\pi) = \mathbb{E}[XY^\top],$$

and the GW objective sees only a second cross-moment. Gaussian marginals are therefore closed under optimal Gaussian replacement, and in the nondegenerate equal-dimensional centered case every optimizer is linear Gaussian.

For squared Euclidean structures, the corresponding matrix acts on the lift

$$(\|x\|^2, x, 1).$$

Equivalently, in the centered case the fixed-point OT cost contains both a bilinear term and the radial product

$$-\|x\|^2 \|y\|^2 .$$

That fourth-order term can encode nonlinear radial dependence while keeping both marginals Gaussian, which is exactly why the Gaussian conclusion fails.